

Ryan Punamiya

CONTACT INFORMATION

Research Scientist @ NVIDIA GEAR Lab
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RESEARCH INTERESTS

My research is centered at the intersection of machine learning, computer, vision and robotics. I am interested by how we can maximize transfer from scalable human data for robot learning. Additionally, I am also interested by data diversity for robotics and generalization in robot foundation models.

RESEARCH EXPERIENCE

Georgia Institute of Technology Atlanta, GA Aug 2023 – Present
Research with Danfei Xu and Judy Hoffman
Domain adaptation from egocentric human video to robots for scalable and generalizable robot learning. Also exploring language steering and vision-language-action alignment within large foundation models for robotics.

EDUCATION

Georgia Institute of Technology Aug 2022 – Present
Bachelor of Science, Computer Science GPA: 4.0
Advised by Danfei Xu and Judy Hoffman

WORK EXPERIENCE

NVIDIA GEAR Lab Santa Clara, CA
Research Scientist June 2026 – Present

Research Intern Feb 2026 – June 2026
Scaling and improving GR00T foundation models with egocentric human data.

CONFERENCE PUBLICATIONS

- * and ** indicate equal contribution; † indicates equal advising.
- [1] **Ryan Punamiya**^{*}, Simar Kareer^{*}, Zeyi Liu, Josh Citron, Ri-Zhao Qiu, Xiongyi Cai, Alexey Gavryushin, Jiaqi Chen, Davide Liconti, Lawrence Y. Zhu, Patcharapong Aphiwetsa, Baoyu Li, Aniketh Cheluva, Pranav Kuppili, Yangcen Liu, Dhruv Patel, Aidan Gao, Hye-Young Chung, Ryan Co, Renee Zbizika, Jeff Liu, Xiaomeng Xu, Haoyu Xiong, Geng Chen, Sebastiano Oliani, Wenkai Xuan, Chenyu Yang, Xi Wang, James Fort, Richard Newcombe, Josh Gao, Jason Chong, Garrett Matsuda, Aseem Doriwala, Marc Pollefeys, Robert Katzschnmann, Xiaolong Wang, Shuran Song, Judy Hoffman, Danfei Xu. “EgoVerse: An Egocentric Human Dataset for Robot Learning from Around the World.” *Robotics: Science and Systems (RSS)*, 2026. Paper | Website.
 - [2] **Ryan Punamiya**, Dhruv Patel, Patcharapong Aphiwetsa, Pranav Kuppili, Lawrence Y. Zhu, Simar Kareer, Judy Hoffman[†], Danfei Xu[†]. “EgoBridge: Domain Adaptation for Generalizable Imitation from Egocentric Human Data.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. Paper | Website.
 - [3] Simar Kareer, Dhruv Patel^{*}, **Ryan Punamiya**^{*}, Pranay Mathur^{*}, Shuo Cheng, Chen Wang, Judy Hoffman[†], Danfei Xu[†]. “EgoMimic: Scaling Imitation Learning via Egocentric Video.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. Paper | Website.

JOURNAL PUBLICATIONS

- [4] Lawrence Y. Zhu, Pranav Kuppili^{*}, **Ryan Punamiya**^{*}, Patcharapong Aphiwetsa^{**}, Dhruv Patel^{**}, Simar Kareer, Sehoon Ha, Danfei Xu. “EMMA: Scaling Mobile Manipulation via Egocentric Human Data.” *IEEE Robotics and Automation Letters (RA-L)*, 2026. Paper | Website.

WORKSHOP PUBLICATIONS	[5] Ryan Punamiya , Dhruv Patel, Patcharapong Aphiwetsa, Pranav Kuppili, Lawrence Y. Zhu, Simar Kareer [†] , Judy Hoffman [†] , Danfei Xu [†] . “EgoBridge: Domain Adaptation for Generalizable Imitation from Egocentric Human Data.” <i>CoRL Human2Robot Workshop</i> , 2025. Oral Presentation; Best Paper Finalist. Website.	
AWARDS	CRA Outstanding Undergraduate Researcher Award Runners Up	2026
	Best Paper Finalist at CoRL Human2Robot Workshop	2025
	President’s Undergraduate Research Award x 3	2024,2025
	Overall Event Winner AI ATL Hackathon (1 of 225)	2023
	Georgia Tech Grand Challenges Grant (\$700)	2023
PROFESSIONAL ACTIVITIES	Organizing	
	Workshop Organizer, Scaling H2R Workshop at CoRL	2026
	Reviewing	
	Conference on Robot Learning Human2Robot Workshop	2025
	Advising	
	Patcharapong Aphiwetsa (BS Georgia Tech)	
	Academic Talks	
	Runway ML	2025
	<i>Human Data as a Scalable Source for Generalizable Robot Learning</i>	